

Problem

The 3N+1 sequence is generated using the following rules: If n is even, the next number is $n / 2$. If n is odd, the next number is $3 \times n + 1$. The sequence continues until it reaches 1. The cycle length of a number is the total number of terms in the sequence, including the starting number and 1.

Given two integers i and j, determine the maximum cycle length among all integers between i and j (inclusive).

If $i > j$, swap them before processing.

Input Format

A single line containing two space-separated integers:

i j

Constraints

$1 \leq i, j \leq 1,000,000$

Output Format

Print three integers:

i j maximum_cycle_length

using the original values of i and j.

Sample Input 0

1 10

Line: 10 Col: 22

Sample Output 0

1 10 20

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Language Python 3



```
1 def cycle(n):
2     count=1
3     while(n!=1):
4         if(n%2==0):
5             n=n//2
6         else:
7             n=3*n+1
8         count+=1
9     return count
10 a,b = map(int,input().split())
11 start=min(a, b)
12 end=max(a, b)
13 maximum=0
14 for n in range(start, end + 1):
15     length=cycle(n)
16     if(length > maximum):
17         maximum=length
18 print(a,b,maximum)
```

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☐ Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?

Test case 0

Test case 1

Test case 2

Compiler Message

Success

